Neural Network in FPGA

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1)Design specifications-

I used Artix-7 -xc7a200 board. So, I made a very basic neural network with 3 layers- input, hidden layers (8) and output layers(3) with and argmax layer. Each neuron computes summation(W*D)+B. This summation is achieved by using accumulator which adds until last is activated. Next there is the hidden layer which calls 8 neurons and above it there is nn_top which encloses the layer and contains the output layer too. It calls layers once and calls 3 separate neurons for output layer and does argmax to give final output.

2)learning outcomes and problems faced-

I learnt to debug using timing diagrams and also learnt when to use blocking and non-blocking assignments.

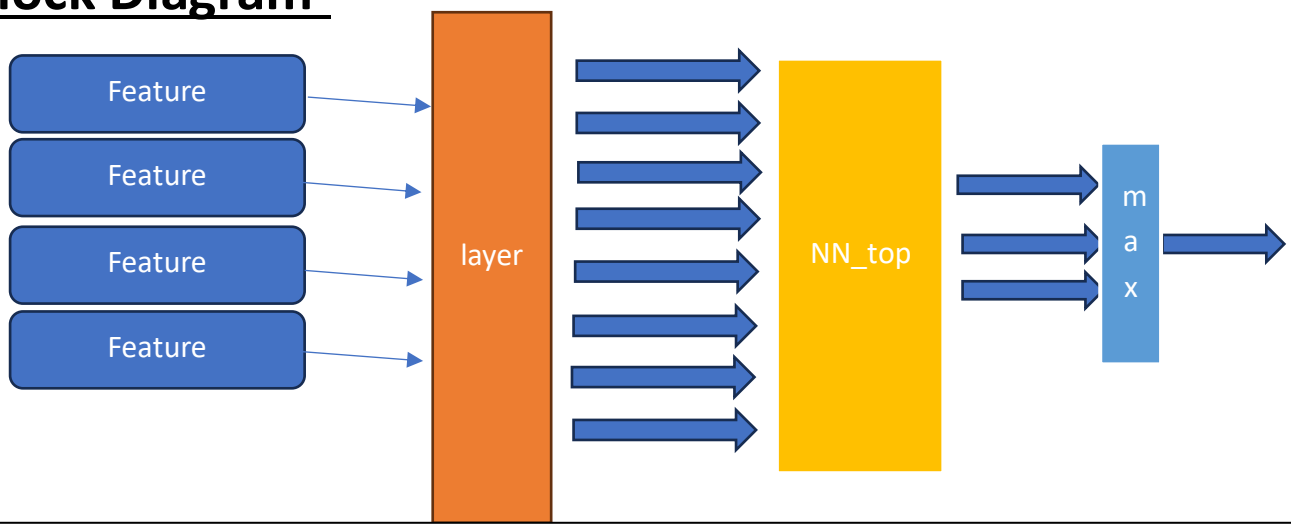
Problem 1- I had to sequentially assign to compute sum then apply ReLU. For this I did blocking assignment and then non-blocking. Since sum is a temporary variable, it does not affect the synthesis.

Problem 2- When last was getting updated, it wasn't adding mult so I had debugged and found it from timing diagram.

Problem 3-In layers, the output was an array, this caused problem when running testbench. So, I had to break it down into multiple outputs.

Problem 4- The toughest problem I faced was in generating bitstream which initially caused errors since .xcd file had only one constraint as per the pdf sent. So, I added in all the constraints I found in the board documentation.

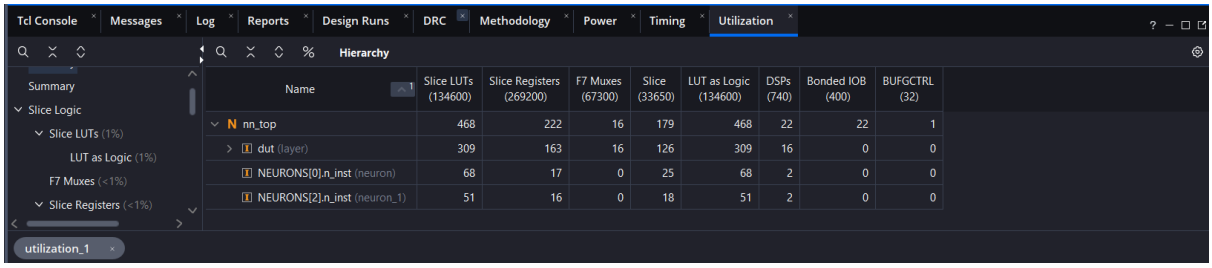
3)Block Diagram-



Input	Hidden layer (8 neurons)	Output layer (3 neurons).

4)Results-

- LUT utilization- It uses around 468 LUTs 1%.



Tcl Console | Messages | Log | Reports | Design Runs | DRC | Methodology | Power | Timing | Utilization

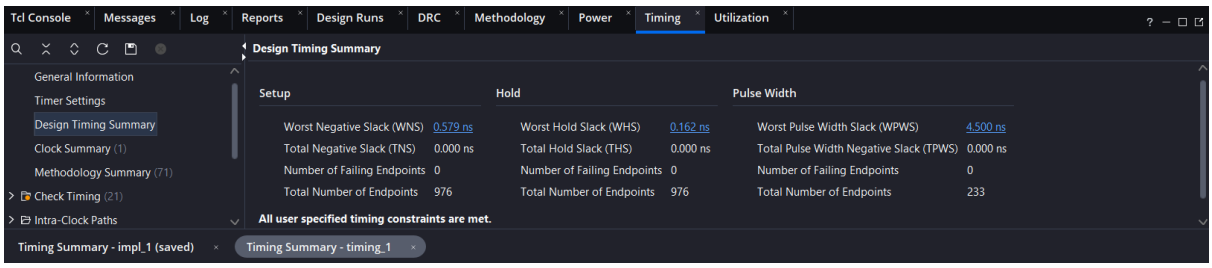
Hierarchy

Summary
Slice Logic
Slice LUTs (1%)
LUT as Logic (1%)
F7 Muxes (<1%)
Slice Registers (<1%)

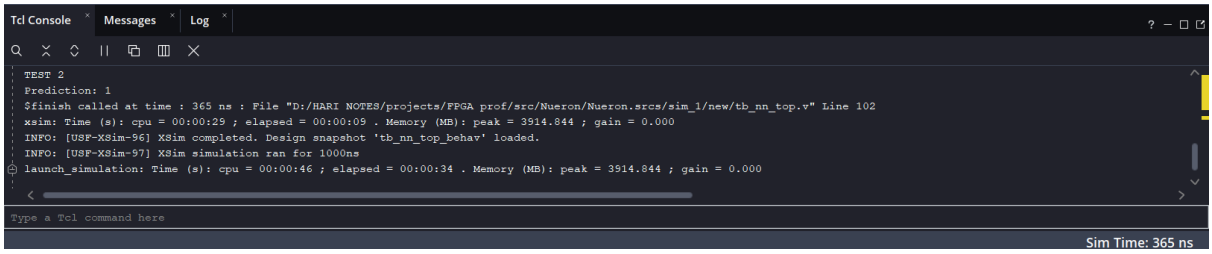
Name	Slice LUTs (134600)	Slice Registers (269200)	F7 Muxes (67300)	Slice (33650)	LUT as Logic (134600)	DSPs (740)	Bonded IOB (400)	BUFGCTRL (32)
nn_top	468	222	16	179	468	22	22	1
dut (layer)	309	163	16	126	309	16	0	0
NEURONS[0].n_inst (neuron)	68	17	0	25	68	2	0	0
NEURONS[2].n_inst (neuron_1)	51	16	0	18	51	2	0	0

utilization_1

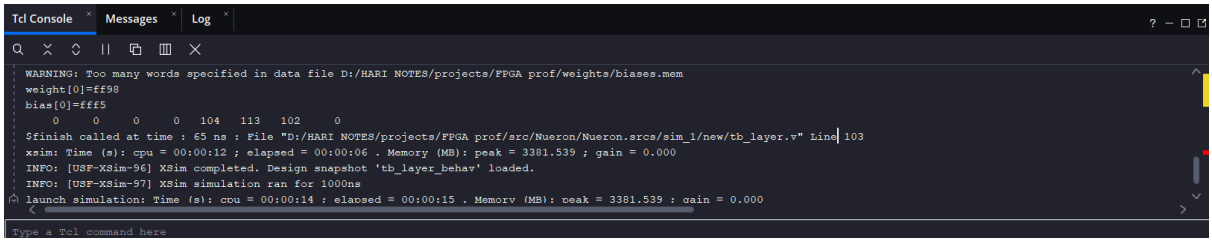
- Time slack- Worst Negative Slack- 0.579ns (it is positive so design meets timing)



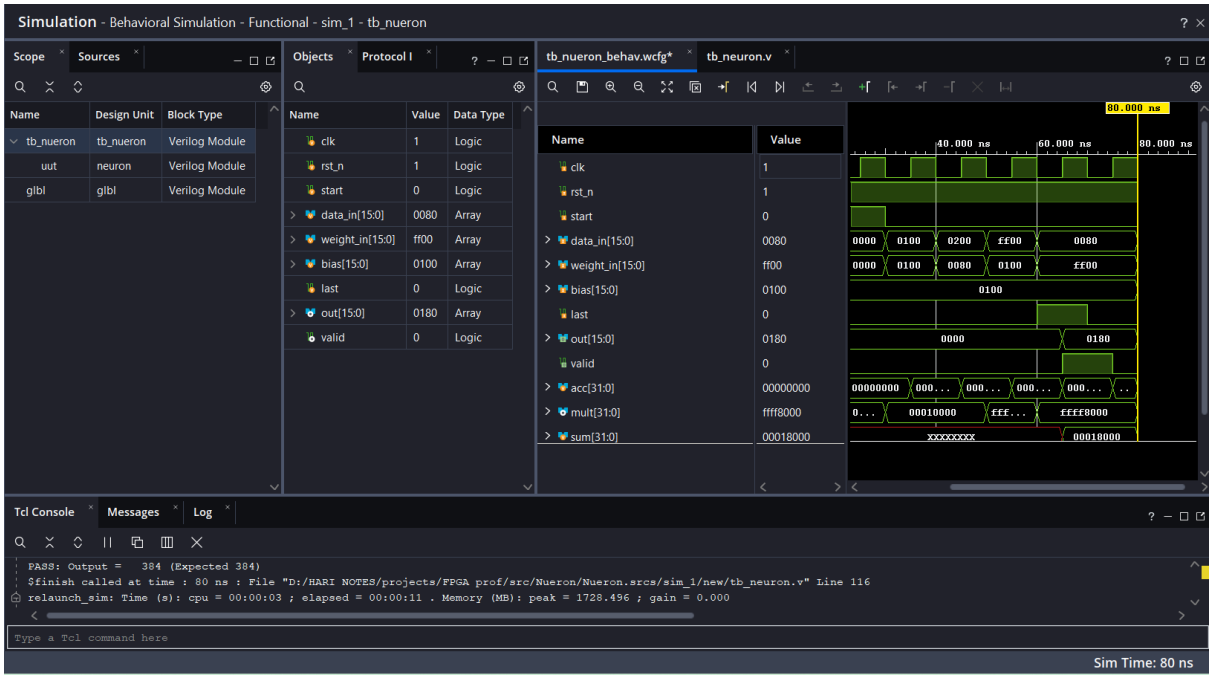
- <u>NN Top sim</u>- It gave successful prediction of 1.



- <u>Layer sim</u>- It successfully gave positive numbers and 0 by ReLU.



- **Neuron sim**- It ran successful giving expected results.



○ Overall info-

Tcl Console | Messages | Log | Reports | Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP	Start
synth_1	constrs_1	synth_design Complete!												649	222	0	0	22	4/11/2
impl_1	constrs_1	route_design Complete!	0.579	0.000	0.162	0.000		0.000	0.169	0	71 Warn			468	222	0	0	22	4/11/2

Synthesis

Status	Complete
Messages	102 critical warnings
	101 warnings
Part	xc7a200tfbg676-2
Strategy	Vivado Synthesis Defaults
Report Strategy	Vivado Synthesis Default Reports
Incremental synthesis	Automatically selected checkpoint

Implementation Summary | Route Status

Status	Complete
Messages	103 critical warnings
	100 warnings
Part	xc7a200tfbg676-2
Strategy	Vivado Implementation Defaults
Report Strategy	Vivado Implementation Default Reports
Incremental implementation	None

DRC Violations

Summary: 2 critical warnings, 110 warnings

Implemented DRC Report

Timing Setup | Hold | Pulse Width

Worst Negative Slack (WNS)	0.579 ns
Total Negative Slack (TNS)	0 ns
Number of Failing Endpoints	0
Total Number of Endpoints	976

Implemented Timing Report

Utilization Post-Synthesis | Post-Implementation

Graph | Table

Power Summary | C

Total On-Chip Power: 0.169 W